

SEC 2-FA4 GROUP 24-VALDEZ, CJ
<https://github.com/ChanceValdez/APM1110/tree/main/FA4>

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Exercise 7.1

Item 2

A binary communication channel carries data as one of two sets of signals denoted by 0 and 1. Owing to noise, a transmitted 0 is sometimes received as a 1, and a transmitted 1 is sometimes received as a 0. For a given channel, it can be assumed that a transmitted 0 is correctly received with probability 0.95, and a transmitted 1 is correctly received with probability 0.75. Also, 70% of all messages are transmitted as a 0.

Let:

- T_0 : Signal 0 is transmitted.
- T_1 : Signal 1 is transmitted.
- R_0 : Signal 0 is received
- R_1 : Signal 1 is received.

We are given:

- $P(T_0) = 0.70 \implies P(T_1) = 1 - 0.70 = 0.30$
- $P(R_0|T_0) = 0.95 \implies P(R_1|T_0) = 1 - 0.95 = 0.05$
- $P(R_1|T_1) = 0.75 \implies P(R_0|T_1) = 1 - 0.75 = 0.25$

If a signal is sent, determine the probability that:

(a) a 1 was received;

For this problem, we use the Law of Total Probability:

$$\begin{aligned} P(R_1) &= P(R_1|T_0)P(T_0) + P(R_1|T_1) \cdot P(T_1) \\ P(R_1) &= (0.05 \cdot 0.70) + (0.75 \cdot 0.30) \\ P(R_1) &= 0.035 + 0.225 = 0.26 \end{aligned}$$

The probability that a 1 was received is 0.26 or 26%.

(b) a 1 was transmitted given that a 1 was received.

For this problem, we use Bayes' Theorem:

$$P(T_1|R_1) = \frac{P(R_1|T_1)P(T_1)}{P(R_1)}$$
$$P(T_1|R_1) = \frac{0.225}{0.26} \approx 0.8654$$

The probability that a 1 was transmitted given that a 1 was received is approximately 0.8654 or 86.54%.

R Code for Verification

```
P_T0 <- 0.70
P_T1 <- 1 - P_T0 # 0.30
P_R0_given_T0 <- 0.95
P_R1_given_T0 <- 1 - P_R0_given_T0 # 0.05
P_R1_given_T1 <- 0.75
P_R0_given_T1 <- 1 - P_R1_given_T1 # 0.25

priors <- c(P_T0, P_T1)
conditionals <- c(P_R1_given_T0, P_R1_given_T1)

P_R1_result <- sum(conditionals * priors)

calc_bayes_posterior <- function(prior, conditional, total_likelihood) {
  posterior <- (conditional * prior) / total_likelihood
  return(posterior)
}

P_T1_given_R1_result <- calc_bayes_posterior(
  prior = P_T1,
  conditional = P_R1_given_T1,
  total_likelihood = P_R1_result
)

cat("The probability that a 1 was received is ", P_R1_result,
    " (or ", P_R1_result * 100, "%)\n", sep = "")

## The probability that a 1 was received is 0.26 (or 26%)

cat("The probability that a 1 was transmitted given that a 1 was received is approximately: \n",
    round(P_T1_given_R1_result, 4), " (or ", round(P_T1_given_R1_result * 100, 2), "%)\n", sep = "")

## The probability that a 1 was transmitted given that a 1 was received is approximately:
## 0.8654 (or 86.54%)
```

Item 7

There are three employees working at an IT company: Jane, Amy, and Ava, doing 10%, 30%, and 60% of the programming, respectively. 8% of Jane's work, 5% of Amy's work, and just 1% of Ava's work is in error.

Step by Step Computation

Let:

- J : Jane's programming.
- Am : Amy's programming.
- Av : Ava's programming.
- E : The program contains an error.

We are given:

- $P(J) = 0.10 \implies P(E|J) = 0.08$
- $P(Am) = 0.30 \implies P(E|Am) = 0.05$
- $P(Av) = 0.60 \implies P(E|Av) = 0.01$

(a) What is the overall percentage of error?

To find the total probability of an error occurring, $P(E)$, we use the Law of Total Probability:

$$\begin{aligned} P(E) &= P(E|J)P(J) + P(E|Am)P(Am) + P(E|Av)P(Av) \\ P(E) &= (0.08 \cdot 0.10) + (0.05 \cdot 0.30) + (0.01 \cdot 0.60) \\ P(E) &= 0.008 + 0.015 + 0.006 \\ P(E) &= 0.029 \end{aligned}$$

The overall percentage of error in the company is 0.029 or 2.9%.

(b) If a program is found with an error, who is the most likely person to have written it?

To determine who is most likely responsible for a given error, we use Bayes' Theorem to calculate the posterior probability for each employee:

$$\begin{aligned} P(J|E) &= \frac{P(E|J)P(J)}{P(E)} = \frac{0.008}{0.029} \approx 0.2759 \text{ (or 27.59\%)} \\ P(Am|E) &= \frac{P(E|Am)P(Am)}{P(E)} = \frac{0.015}{0.029} \approx 0.5172 \text{ (or 51.72\%)} \\ P(Av|E) &= \frac{P(E|Av)P(Av)}{P(E)} = \frac{0.006}{0.029} \approx 0.2069 \text{ (or 20.69\%)} \end{aligned}$$

When comparing the posterior probabilities ($0.5172 > 0.2759 > 0.2069$), Amy is the most likely person to have written the program containing the error.

R Code for Verification

```
work_share <- c(Jane = 0.10, Amy = 0.30, Ava = 0.60)
error_rates <- c(Jane = 0.08, Amy = 0.05, Ava = 0.01)
error_contributions <- work_share * error_rates

total_error <- sum(error_contributions)
cat("The overall percentage of error:", total_error * 100, "%\n")
```

```
## The overall percentage of error: 2.9 %
```

```

calc_bayes_posterior <- function(prior, conditional, total_likelihood) {
  posterior <- (conditional * prior) / total_likelihood
  return(posterior)
}

posterior_probs <- calc_bayes_posterior(
  prior = work_share,
  conditional = error_rates,
  total_likelihood = total_error
)

cat("Probability that a Person is Responsible for a Given Error:\n")

```

```
## Probability that a Person is Responsible for a Given Error:
```

```
print(round(posterior_probs * 100, 2))
```

```
## Jane Amy Ava
## 27.59 51.72 20.69
```

```

most_likely <- names(which.max(posterior_probs))
cat("Most Likely Person to have Written the Error:", most_likely, "\n")

```

```
## Most Likely Person to have Written the Error: Amy
```

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